DL TUMOR CLASSIFICATION

40-Image In-the-Wild Web MRI Stress Test

Consolidated methodology, integrity audit, results, analysis, and transfer record

40 source-verified images	**10 per class** balanced cohort	**15** frozen pipelines	**600** model-image predictions

Executive finding

The frozen pipelines achieved only 25.0%-42.5% accuracy on this deliberately heterogeneous web-sourced MRI set, despite internal validation around 94.5%-99.4%. The dominant failure pattern was a strong shift toward the **notumor** class, with especially low recall for glioma and pituitary. This experiment therefore demonstrates substantial sensitivity to out-of-distribution MRI presentation.

Purpose of this PDF

This standalone record documents exactly what was done for the 40-image web stress test, why each step was necessary, how image overlap was controlled before inference, what the 15 frozen models produced, and what can and cannot be concluded. It is designed for personal reference and for transfer into future ChatGPT/Claude work without reconstructing the experiment from chat history.

Current status: WEB STRESS TEST COMPLETE

Source collection, pre-inference overlap audit, visual review, replacement of one reused image, final clean audit, frozen 15-model evaluation, and scientific interpretation are complete. No per-image consensus appendix is required for the core project.

1. Why this experiment was added

The structured BDNeuro external validation had already shown that the project generalized less strongly outside the development dataset than internal validation suggested. However, the original guide question was more practical: *what happens if an MRI image comes from outside the training dataset, such as a source found on the web?*

A formal external dataset and an in-the-wild web image are not the same problem. The web setting can introduce much stronger distribution shift through image formatting, sequence differences, field strength, orientation, crop, publication export, lesion subtype, and other acquisition/presentation differences. The 40-image experiment was therefore designed as a small **stress test**, not another tuning set and not a clinical validation study.

Locked purpose
Measure how the already-trained, already-frozen pipelines behave on a small, source-verified, heterogeneous set of MRI images that were not intentionally drawn from the development corpus. The aim is robustness probing, not model selection.

Why 40 images?

- Four classes were preserved: glioma, meningioma, notumor, pituitary.

- Ten images per class keeps the stress test exactly balanced and easy to audit manually.

- With 40 images, one case equals 2.5 percentage points, so small numerical differences must not be overinterpreted.

- The cohort is intentionally small enough for source verification and visual integrity review before any prediction is viewed.

Relationship to the rest of the project

Evidence level	Purpose	Status
Internal train/validation/test	Development and historical benchmark	Complete
Material-clean internal test	Post-hoc leakage-aware internal check	Complete
BDNeuro exact-overlap-clean	Structured external validation	Complete
40-image web set	Exploratory in-the-wild robustness stress test	Complete

2. Building the 40-image source list

Before model inference, a fixed candidate list was built with exactly 10 source-verified images per class. The source list was stored in `Web_Stress_Test_40_Source_List.xlsx`.

Rule	What we did / why
Class balance	10 glioma, 10 meningioma, 10 notumor, 10 pituitary. This prevents class prevalence from driving the basic accuracy result.
Source verification	The diagnosis/normal status had to be supported by the original medical article or image page rather than inferred from visual appearance alone.
Source type	Tumor images were drawn mainly from open-access case reports/medical literature; normal images came from explicitly normal or healthy-volunteer MRI sources.
Panel selection	For multi-panel figures, only the specified MRI panel was isolated. Cropping was allowed to isolate the panel, but image content was not deliberately enhanced or modified.
Pre-inference replacement	Images could be rejected/replaced only before model predictions were seen, for reasons such as ambiguous labeling, unsuitable presentation, or confirmed overlap.
Post-inference rule	After predictions were viewed, the cohort was not to be modified in response to failures. This prevents the stress test from becoming an informal tuning set.

Initial directory structure

`external_data/web_stress_40_raw/{glioma, meningioma, notumor, pituitary}`

The collection check returned exactly 10 images in each class, for 40 total.

Important interpretation
Source verification establishes the intended class label for this exploratory test. It does not make the set clinically standardized, patient-independent, or representative of a target population.

3. Pre-inference overlap and integrity audit

No model was evaluated until the 40-image set had been screened against both prior image pools. The audit script was `actual_training_scripts/audit_web_stress_40.py`.

Audit component	Reference / decision rule
Within-web duplicate check	Exact file-byte and exact decoded-pixel duplicate checks among the 40 images.
Nickparvar reference	All 7,023 original development images.
BDNeuro reference	The 2,644-image exact-overlap-clean BDNeuro cohort.
Exact file hash	SHA-256 of file bytes.
Exact decoded-pixel hash	Objective pixel-identity check after decoding to RGB; exact match is exclusion evidence.
pHash	64-bit perceptual hash used only as a similarity-review signal. pHash similarity alone was not automatic exclusion evidence.
Review thresholds	pHash <= 6 generated candidates; <= 4 was treated as a stronger review flag.

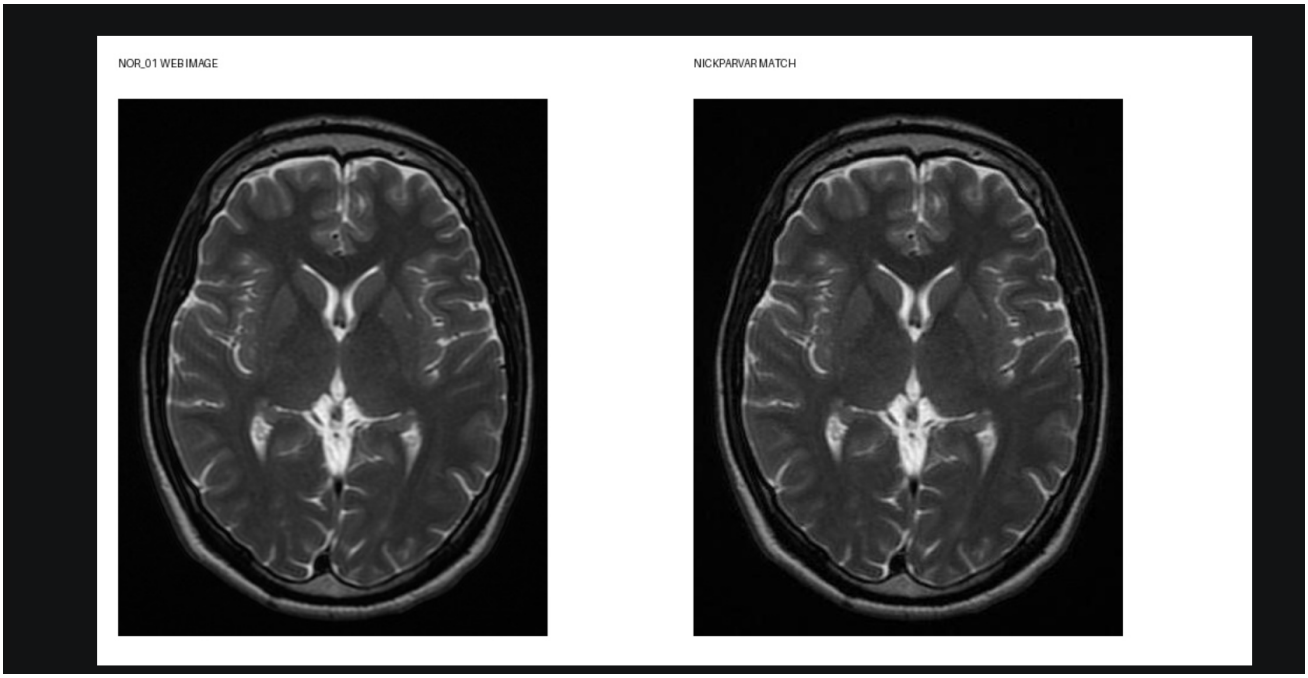
First audit outcome

Measure	Result
Within-web exact file pairs	0
Within-web exact pixel pairs	0
Cross-dataset exact file pairs	0
Cross-dataset exact pixel pairs	0
Unique web images with exact pixel overlap	0
Unique web images with pHash <= 6	2
Unique web images with pHash <= 4	1
Best-match distances	One image at distance 0; one image at distance 6

The two perceptual-hash candidates were therefore reviewed visually before any model prediction was accepted as part of the experiment.

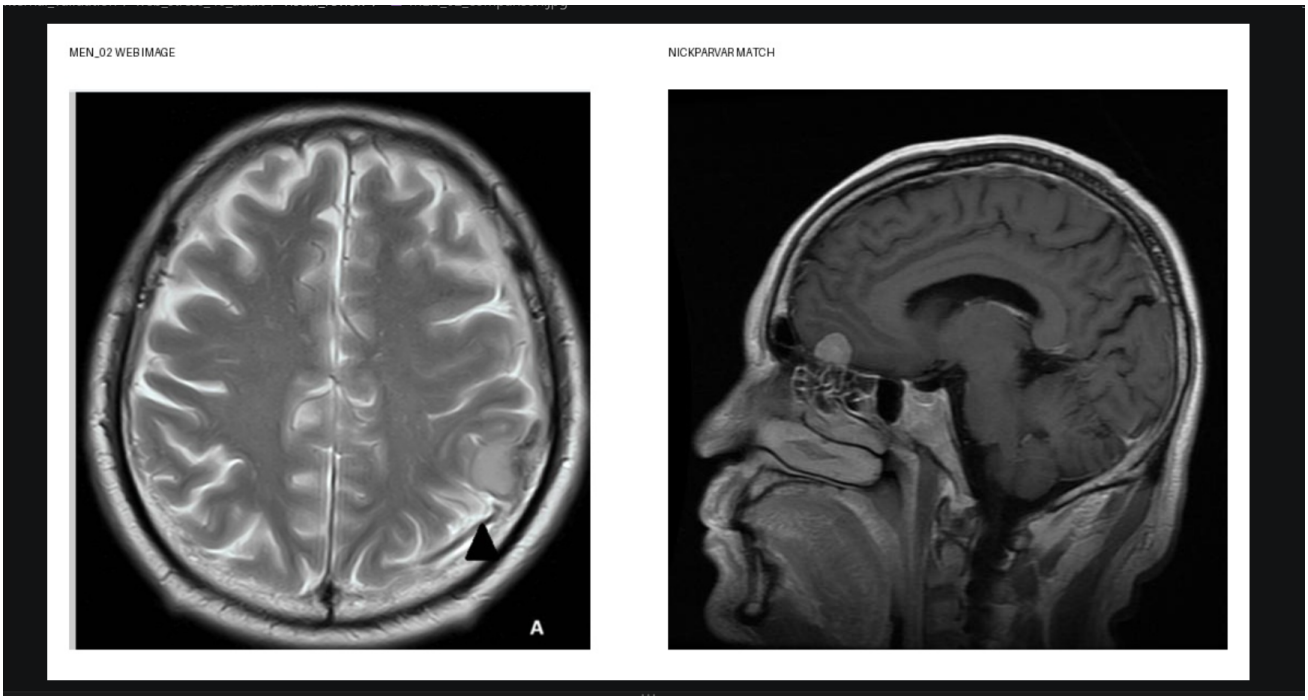
4. Visual review of the two pHash candidates

NOR_01 - pHash distance 0 against Nickparvar Testing/notumor/Te-no_0227.jpg



Visual review showed the same underlying axial T2 brain slice: the ventricular shape, cortical contours, sulci, midline structures, and slice level matched. Although the files were not exact byte/pixel duplicates, the perceptual-hash flag correctly identified reused image content. **Decision: reject and replace NOR_01 before inference.**

MEN_02 - pHash distance 6 against Nickparvar Training/meningioma/Tr-me_0633.jpg



The images were clearly different: the web image and Nickparvar candidate differed in orientation, anatomy, lesion appearance, and slice view. The pHash-6 match was therefore treated as a false-positive similarity signal. **Decision: retain MEN_02.**

5. Replacement, final audit, and cohort lock

The original NOR_01 was deleted from the active stress-test folder and replaced with a new source-verified normal MRI image. The entire audit was then rerun from the beginning.

Final audit measure	Result
Web images	40
Class counts	10 / 10 / 10 / 10
Within-web exact file duplicates	0
Within-web exact pixel duplicates	0
Cross-dataset exact file duplicates	0
Cross-dataset exact pixel duplicates	0
Unique web images with exact pixel overlap	0
pHash <= 6	1 (MEN_02, already visually reviewed)
pHash <= 4	0

Lock decision
After the replacement passed the objective overlap audit, the 40-image set was treated as frozen for inference. No image was to be removed, recropped, replaced, or substituted after model outputs were viewed.

Evaluator-recorded cohort identity

The external evaluator recorded the final dataset fingerprint:

```
26da5c49d7a45c27ea4fbbf2a44957ffafba1cede0dd85c29a3510dc42f9591c
```

This fingerprint, together with the 10-per-class counts in the result package, provides a reproducible identity check for the evaluated cohort.

6. Frozen 15-model evaluation protocol

The same 15 pipelines that had already been frozen before BDNeuro evaluation were reused. No model was selected, retrained, refitted, calibrated, or tuned using the web results.

Stage	Per-backbone representative	Count
Phase 1	Early-stopping pipeline	5
Phase 2A	Standardized 30-epoch representative	5
Phase 2B	Preselected tuned representative	5
Total	Five backbones x three stages	15

The backbones remained VGG16, VGG19, ResNet50, ResNet152, and EfficientNet-B3. Each frozen pipeline used its saved CNN-LSTM checkpoint and its saved StandardScaler, PCA, and SVM artifacts. The web set was inference-only.

Scale of the experiment
15 frozen pipelines x 40 locked images = 600 predictions. Because the cohort is balanced, overall accuracy equals macro recall. Weighted F1 also equals macro F1 because each class has identical support.

Output location

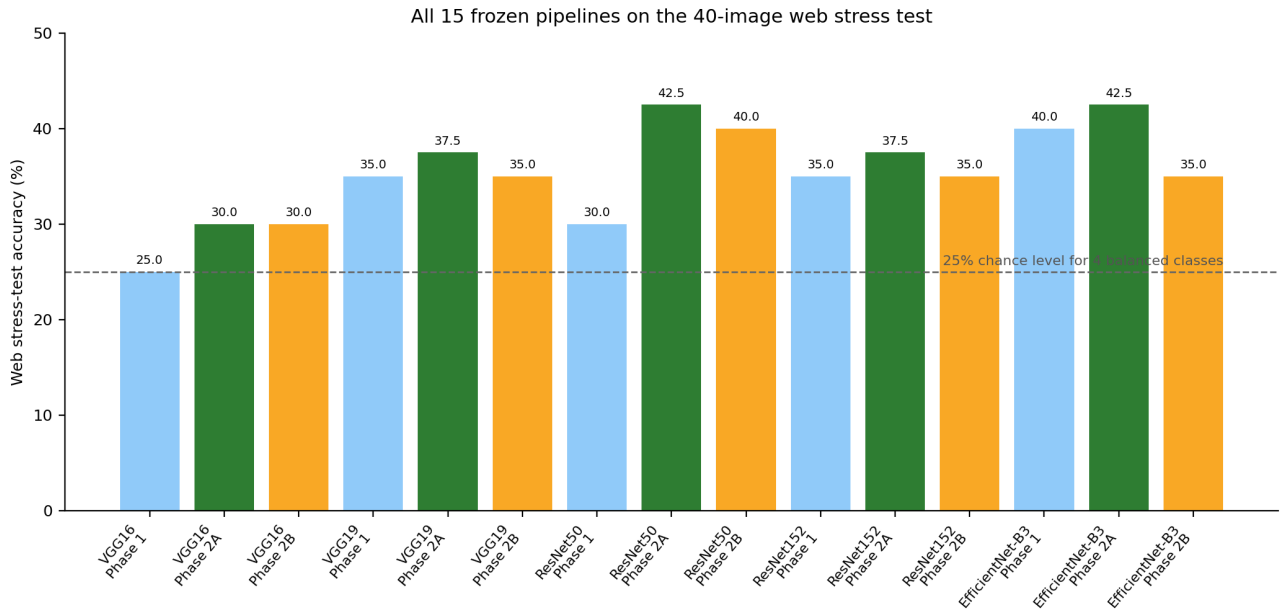
`phase2/external_validation/comparison_15_models_web_stress_40/`

Master artifacts: `external_comparison_15.csv` and `external_comparison_15.json`.

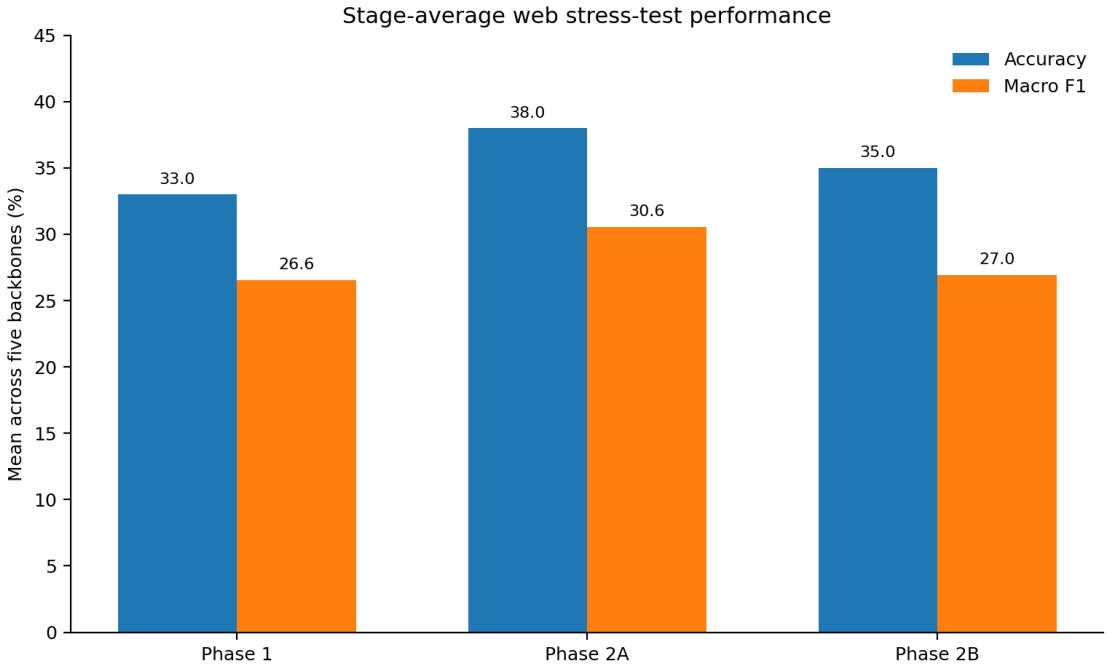
7. Complete 15-pipeline web stress-test results

Backbone	Stage	Variant	Internal val acc	Web acc	Macro F1	Gap (pp)
VGG16	Phase 1	early stop	94.93%	25.0%	14.58%	-69.93
VGG16	Phase 2A	baseline	95.89%	30.0%	23.23%	-65.89
VGG16	Phase 2B	lr_5e-4	96.50%	30.0%	21.59%	-66.50
VGG19	Phase 1	early stop	94.49%	35.0%	28.89%	-59.49
VGG19	Phase 2A	baseline	95.63%	37.5%	27.78%	-58.13
VGG19	Phase 2B	dropout_0.5	95.54%	35.0%	25.45%	-60.54
ResNet50	Phase 1	early stop	97.55%	30.0%	26.49%	-67.55
ResNet50	Phase 2A	baseline	98.69%	42.5%	38.18%	-56.19
ResNet50	Phase 2B	dropout_0.2	99.13%	40.0%	33.91%	-59.13
ResNet152	Phase 1	early stop	98.25%	35.0%	28.08%	-63.25
ResNet152	Phase 2A	baseline	98.60%	37.5%	31.66%	-61.10
ResNet152	Phase 2B	lstm_512	99.30%	35.0%	25.74%	-64.30
EfficientNet-B3	Phase 1	early stop	98.51%	40.0%	34.78%	-58.51
EfficientNet-B3	Phase 2A	baseline	99.30%	42.5%	31.97%	-56.80
EfficientNet-B3	Phase 2B	lstm_512	99.39%	35.0%	28.07%	-64.39

Highest observed accuracy
42.5% (17/40 correct), achieved by both ResNet50 Phase 2A and EfficientNet-B3 Phase 2A. Their macro F1 values differed: 38.18% vs 31.97%, indicating different class-balance behavior despite identical total correct counts.



8. Phase-level pattern: Phase 2A generalized best on average

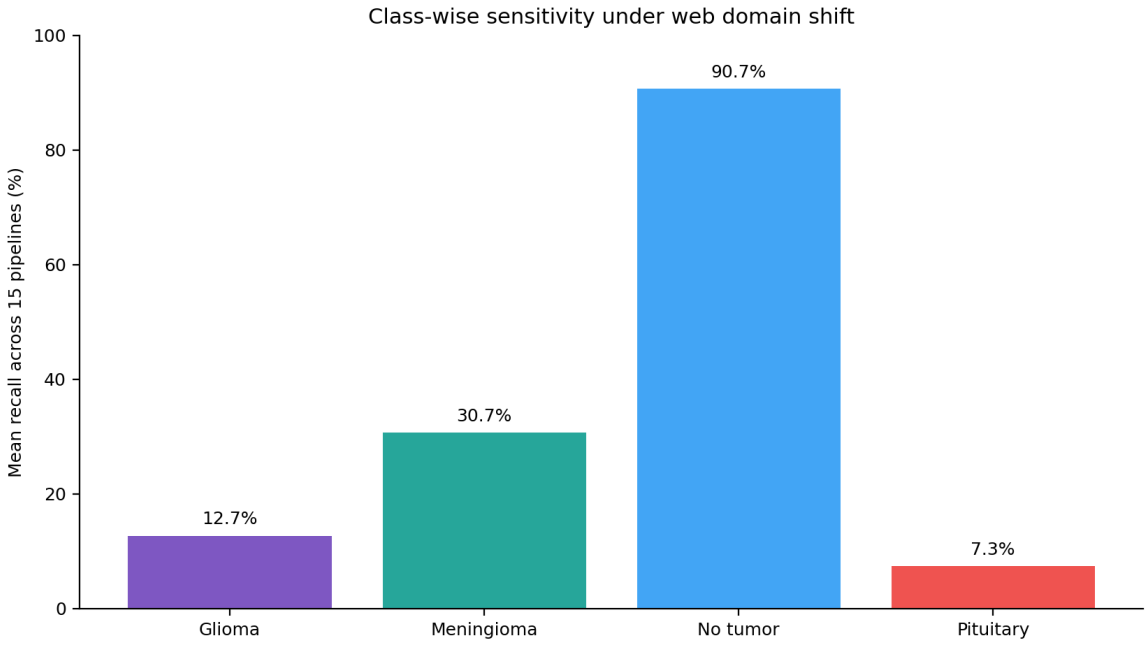


Stage	Mean accuracy	Mean macro F1
Phase 1	33.0%	26.56%
Phase 2A	38.0%	30.56%
Phase 2B	35.0%	26.95%

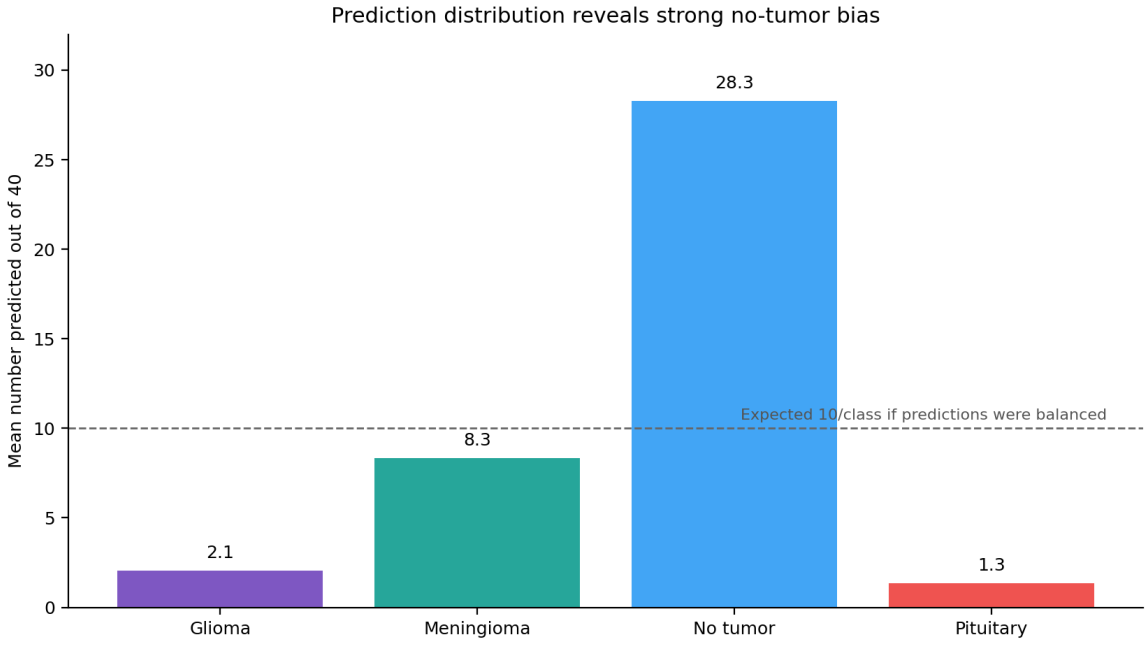
Phase 2A again showed the strongest average external behavior, consistent with the earlier BDNeuro pattern. Phase 2B hyperparameter tuning did not reliably transfer to the web domain. Examples: ResNet50 improved from 30.0% in Phase 1 to 42.5% in Phase 2A and 40.0% in Phase 2B; EfficientNet-B3 fell from 42.5% in Phase 2A to 35.0% in Phase 2B; ResNet152 Phase 2B returned to 35.0% after 37.5% in Phase 2A.

This does **not** establish Phase 2A as a universally superior training protocol. It shows that the internally selected Phase 2B changes were not consistently robust under this stronger distribution shift.

9. Main failure mode: prediction collapse toward no tumor



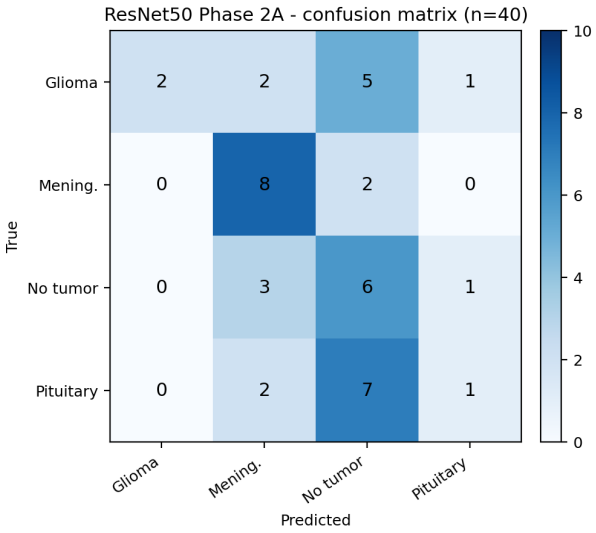
True class	Mean recall across 15
Glioma	12.7%
Meningioma	30.7%
No tumor	90.7%
Pituitary	7.3%

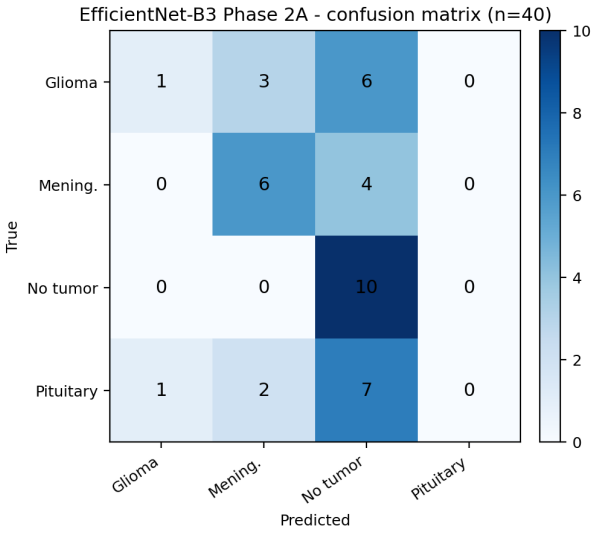


Across the 15 pipelines, an average model predicted approximately **28.3 of the 40 images as notumor**, compared with 8.3 as meningioma, 2.1 as glioma, and 1.3 as pituitary. This is not random guessing. It is a systematic shift in decision behavior under the web-image domain.

Core robustness finding: The models retained high sensitivity to normal cases but frequently mapped heterogeneous tumor images to the no-tumor class. Glioma and pituitary were especially vulnerable, with mean recalls of only about 12.7% and 7.3% respectively.

10. Confusion matrices for the two 42.5% pipelines



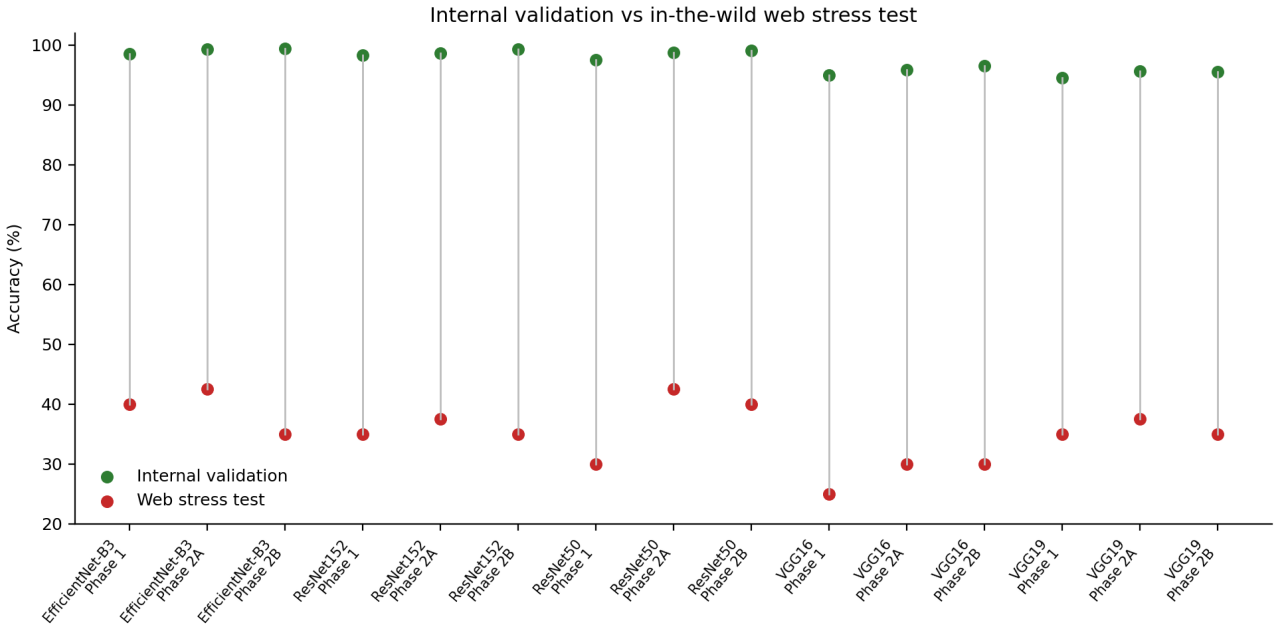


ResNet50 Phase 2A: 2/10 glioma, 8/10 meningioma, 6/10 notumor, and 1/10 pituitary were correct. Its 42.5% accuracy was therefore supported by a comparatively broader spread of correct tumor predictions, yielding macro F1 38.18%.

EfficientNet-B3 Phase 2A: 1/10 glioma, 6/10 meningioma, 10/10 notumor, and 0/10 pituitary were correct. It reached the same 17/40 correct count, but the concentration of success in the no-tumor class reduced macro F1 to 31.97%.

This illustrates why accuracy alone is insufficient for this balanced four-class problem: two pipelines can have identical accuracy while behaving very differently across clinically meaningful categories.

11. Internal validation did not predict web robustness

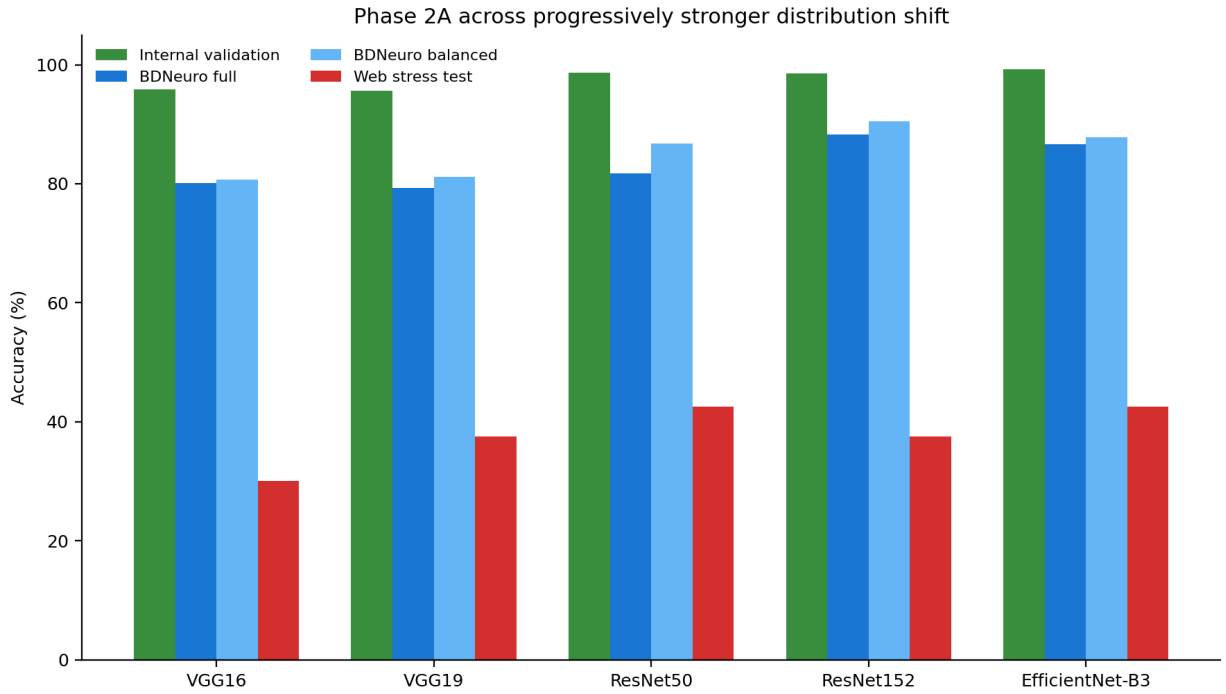


Internal validation accuracy for the 15 frozen candidates ranged from approximately 94.5% to 99.4%, but web accuracy ranged from 25.0% to 42.5%. The internal-to-web drops therefore ranged from roughly 56 to 70 percentage points.

The largest observed drop was VGG16 Phase 1: 94.93% internal validation to 25.0% web accuracy (-69.93 pp). Even the two 42.5% web pipelines fell by more than 56 percentage points relative to their internal validation values.

Interpretation
High performance on the saved internal split - even when reinforced by strong structured external performance - does not imply robustness to unrestricted presentation shift. The web experiment exposed a substantially harder domain.

12. Structured external validation vs web stress testing



Backbone	Internal val	BDNeuro full	BDNeuro balanced	Web 40
VGG16	95.89%	80.07%	80.70%	30.0%
VGG19	95.63%	79.27%	81.13%	37.5%
ResNet50	98.69%	81.73%	86.77%	42.5%
ResNet152	98.60%	88.31%	90.47%	37.5%
EfficientNet-B3	99.30%	86.65%	87.86%	42.5%

This sequence of experiments answers progressively harder questions. BDNeuro asks whether the model transfers to another structured four-class MRI dataset after exact cross-dataset overlap is removed. The web test asks whether the same frozen pipeline remains robust to a much less standardized collection drawn from unrelated medical sources.

The much lower web scores are therefore evidence of **distribution-shift sensitivity**, not evidence that the BDNeuro evaluation was invalid. The two experiments measure different generalization regimes.

13. Why the web stress test was much harder

The source list deliberately included heterogeneous MRI presentations rather than reproducing the visual style of the development dataset. Recorded source metadata included different sequence types and atypical presentations, and the images were often publication-derived panels rather than standardized dataset exports.

Source of shift	Why it matters
MRI sequence variation	T1, contrast-enhanced T1, T2, FLAIR, SWI and other acquisition styles can change tissue contrast and lesion appearance.
Orientation and slice level	Axial predominated, but source material can differ in viewpoint and framing; even axial slices can represent very different anatomical levels.
Tumor phenotype diversity	Glioblastoma, oligodendroglioma variants, unusual meningioma locations, macro/microadenomas and complicated pituitary presentations expand beyond a narrow visual prototype.
Publication processing	Figures may be cropped, rescaled, compressed, annotated, or exported differently from original scanner data.
Healthy-volunteer research imaging	Normal images can come from unusual research protocols/field strengths rather than routine clinical acquisition.
Small convenience sample	The set intentionally probes robustness but is not sampled to estimate population-level clinical performance.

What the experiment demonstrates

The deployed image classifier is strongly dependent on the development-domain visual distribution. Robustness decreases as acquisition and presentation diverge from that domain, especially for tumor classes.

14. Scientific interpretation rules and limitations

Do / say	Do not say
Call it a "40-image source-verified in-the-wild web MRI stress test."	Do not call it a second formal clinical external validation cohort.
Report descriptive accuracy, macro F1, class recall and confusion patterns.	Do not claim 42.5% is the clinical accuracy of the system.
State that the set was balanced 10/class and one image = 2.5 pp.	Do not overrank pipelines based on one- or two-image differences.
State that exact cross-dataset overlap was zero after replacement.	Do not claim patient-level independence; public web images do not provide reliable patient IDs.
State that one pHash-0 reused normal image was discovered and replaced before inference.	Do not use pHash similarity alone as proof of duplication.
State that MEN_02 pHash-6 was visually confirmed different and retained.	Do not remove hard images simply because they resemble prior examples.
Keep the 15 pipelines frozen; no tuning was performed on the web set.	Do not now retrain/select hyperparameters based on these 600 predictions if the set is still described as a held-out stress test.
Describe the strong no-tumor bias and low glioma/pituitary recall.	Do not imply saliency or classification results establish clinical safety or diagnostic utility.

Additional limitations

- Only 40 images were evaluated; this is too small for stable population-level performance estimation.

- The cohort was manually source-curated and is therefore a convenience sample.

- Source labels come from publications/pages, but image-level pathology confirmation and patient-level independence are not uniformly available.

- The image collection intentionally mixes presentation styles; the resulting score should not be compared directly with a standardized dataset score as if the sampling process were identical.

- No repeated-seed or confidence-interval analysis was designed for this small stress-test cohort.

15. Final conclusion and handoff block

Report-ready conclusion

A 40-image, source-verified and class-balanced in-the-wild MRI stress test was constructed to probe robustness beyond structured benchmark datasets. Before inference, all images were audited against the 7,023-image Nickparvar development corpus and the 2,644-image BDNeuro exact-clean cohort. The first audit identified two perceptual-hash candidates: a pHash-0 normal image was visually confirmed to reuse the same underlying Nickparvar MRI and was replaced, while a pHash-6 meningioma image was visually confirmed to be different and retained. The replacement cohort then passed with zero exact file or decoded-pixel overlaps and no pHash candidate <=4. The same 15 pre-frozen CNN-LSTM-PCA-SVM pipelines were evaluated without retraining or refitting, generating 600 predictions. Accuracy ranged from 25.0% to 42.5%, with a strong bias toward the no-tumor class and particularly poor recall for glioma and pituitary. These findings show substantial sensitivity to strong out-of-distribution MRI presentation and demonstrate that high internal or structured-external performance does not imply robustness to heterogeneous web-sourced images.

Compact transfer block

WEB STRESS TEST STATUS 2026-10-03: COMPLETE. Cohort = 40 source-verified MRI images, exactly 10 each glioma/meningioma/notumor/pituitary. Pre-inference audit vs Nickparvar 7,023 + BDNeuro exact-clean 2,644 used exact file hash, decoded-pixel hash and pHash screening. Initial audit: 0 exact overlaps, 2 pHash<=6 candidates. NOR_01 had pHash distance 0 to Nickparvar Te-no_0227 and visual review confirmed the same underlying MRI; it was replaced before inference. MEN_02 had pHash distance 6 to Tr-me_0633 but visual review showed a clearly different MRI and it was retained. Final audit after replacement: 0 within-web exact duplicates, 0 cross-dataset exact file/pixel duplicates, 1 pHash<=6 candidate (MEN_02), 0 pHash<=4. Evaluator dataset fingerprint = 26da5c49d7a45c27ea4fbbf2a44957ffafba1cede0dd85c29a3510dc42f9591c. Same 15 frozen pipelines used (5 Phase1 + 5 Phase2A + 5 Phase2B), no tuning/refitting, 600 predictions. Web accuracy range 25.0%-42.5%. Highest accuracy = ResNet50 Phase2A 42.5% macro-F1 38.18% and EfficientNet-B3 Phase2A 42.5% macro-F1 31.97%. Stage averages: Phase1 33.0% acc / 26.56% macro-F1; Phase2A 38.0% / 30.56%; Phase2B 35.0% / 26.95%. Mean class recall across all 15 = glioma 12.7%, meningioma 30.7%, notumor 90.7%, pituitary 7.3%. Mean predicted class counts out of 40 = glioma 2.1, meningioma 8.3, notumor 28.3, pituitary 1.3. Interpret as exploratory in-the-wild distribution-shift stress test, not clinical accuracy or formal population validation.

Key artifacts and paths

Artifact	Purpose
Web_Stress_Test_40_Source_List.xlsx	Source IDs, source pages, diagnosis/sequence notes and collection instructions.
actual_training_scripts/ audit_web_stress_40.py	Pre-inference exact/pHash integrity audit.
phase2/external_validation/ web_stress_40_audit/	Audit summary, exact-overlap CSVs, pHash candidates and fingerprint manifest.
external_data/ web_stress_40_raw/	Final evaluated 40-image class-folder cohort.
phase2/external_validation/ frozen_15_manifest.json	Pre-existing frozen 15-pipeline selection manifest.
phase2/external_validation/ comparison_15_models_web_stress_40/ external_comparison_15.csv	Master 15-model web results.
phase2/external_validation/ comparison_15_models_web_stress_40/ external_comparison_15.json	Full metrics, confusion matrices and per-class classification reports.

Project implication: The web stress test is complete. Per-image consensus analysis is optional and not necessary for the core project. The next practical step is final documentation/report/PPT integration rather than further model tuning on this 40-image set.